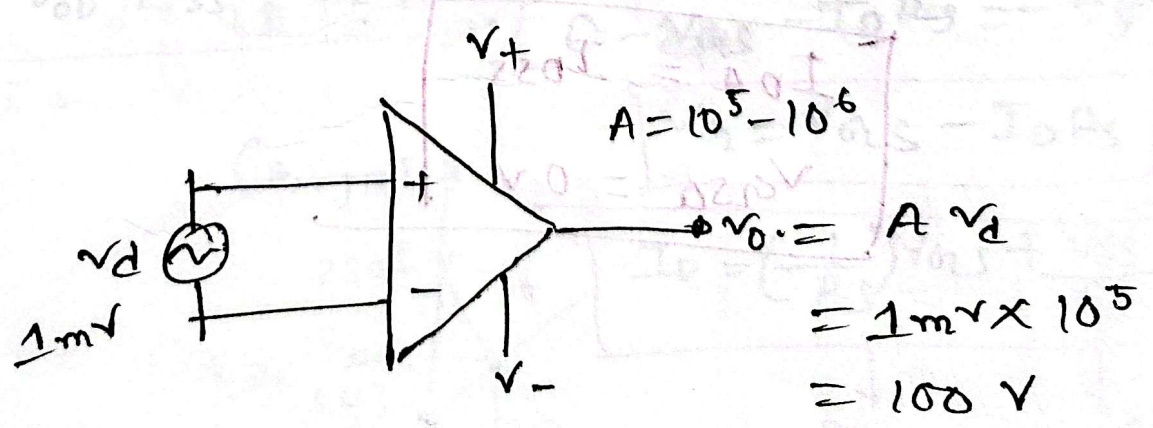
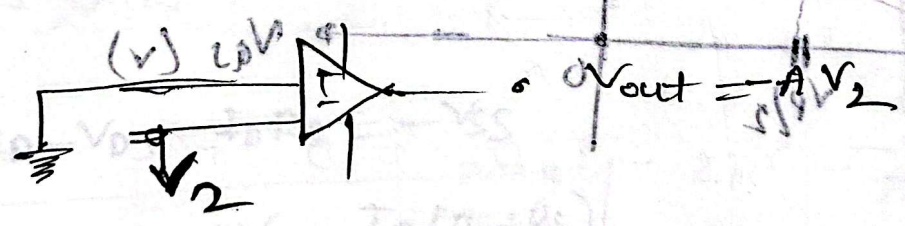
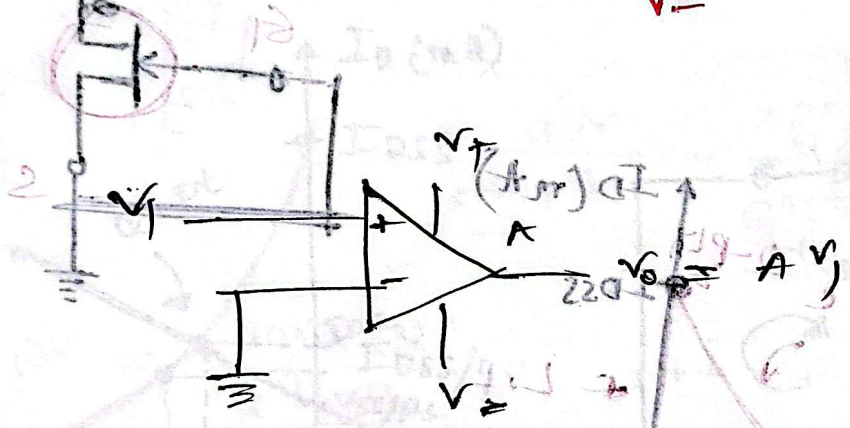
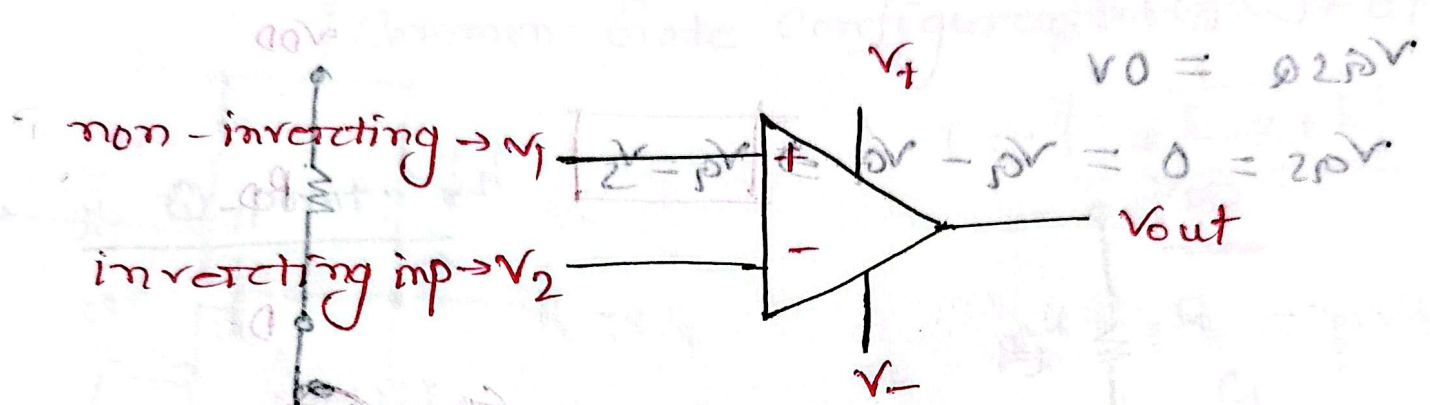


Op-amps

$$(V_{DS} < V_{DD}) \Rightarrow +9-0$$

$$V_0 = 0.2V$$



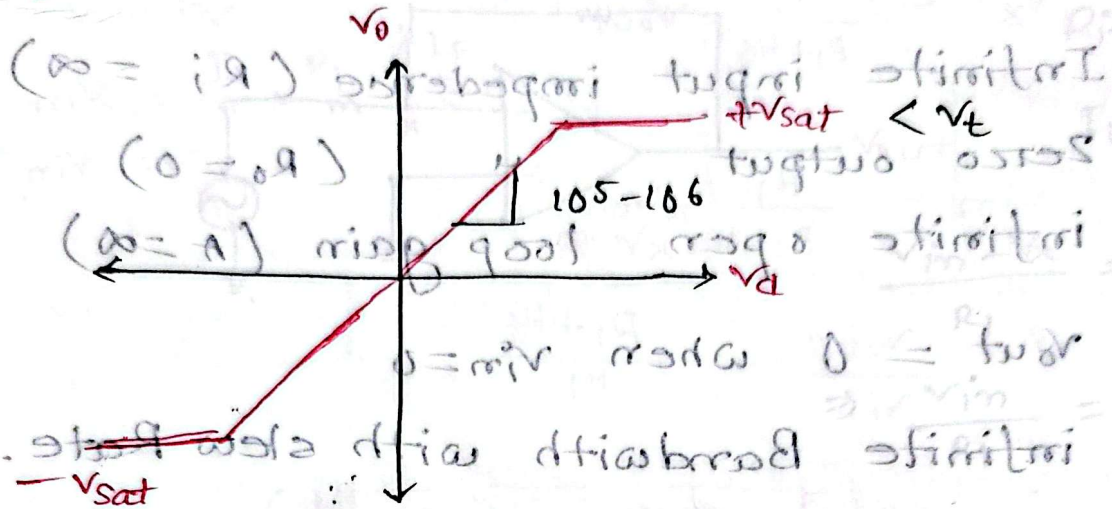
$$A = 10^5 - 10^6$$

$$V_0 = A V_2$$

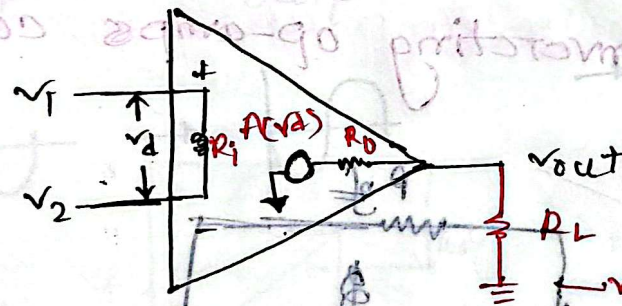
$$= 1mV \times 10^5$$

$$= 100V$$

Voltage transfer curve: -90° loop



Equivalent circuit



$$R_i = \infty$$

$$R_o = 0$$

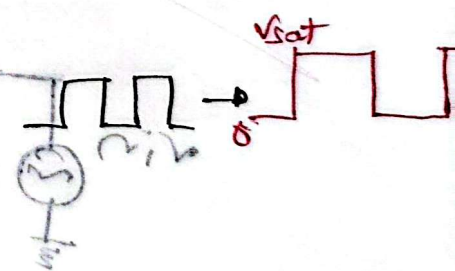
$$B.W = \infty$$

$$A = \infty$$

bandwidth

slew rate $= \infty$

$$A_{MRR} = \frac{A_d}{A_c} = \infty$$



Ideal Op-amps Characteristics

Infinite input impedance ($R_i = \infty$)

Zero output impedance ($R_o = 0$)

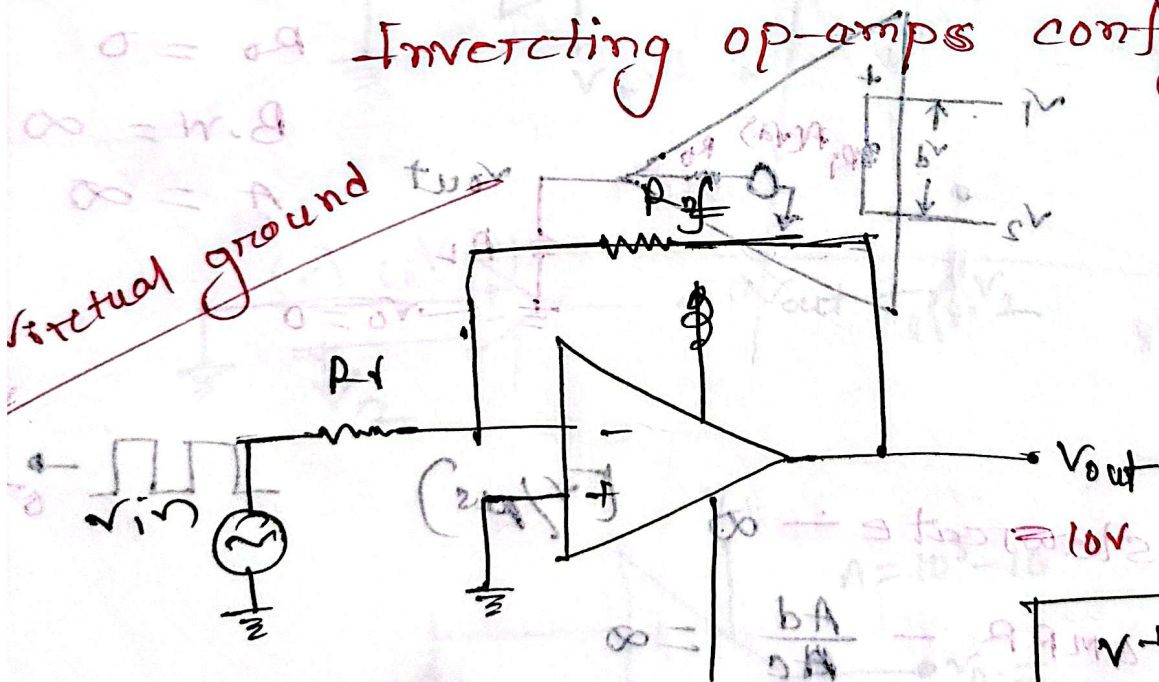
Infinite open loop gain ($A = \infty$)

$V_{out} = 0$ when $V_{in} = 0$

Infinite Bandwidth with slew Rate.

Infinite CMRR (Common mode rejection ratio)

Inverting op-amps configuration



$$A_{OL} = 10^6$$

$$V_{out} = A \times V_d$$

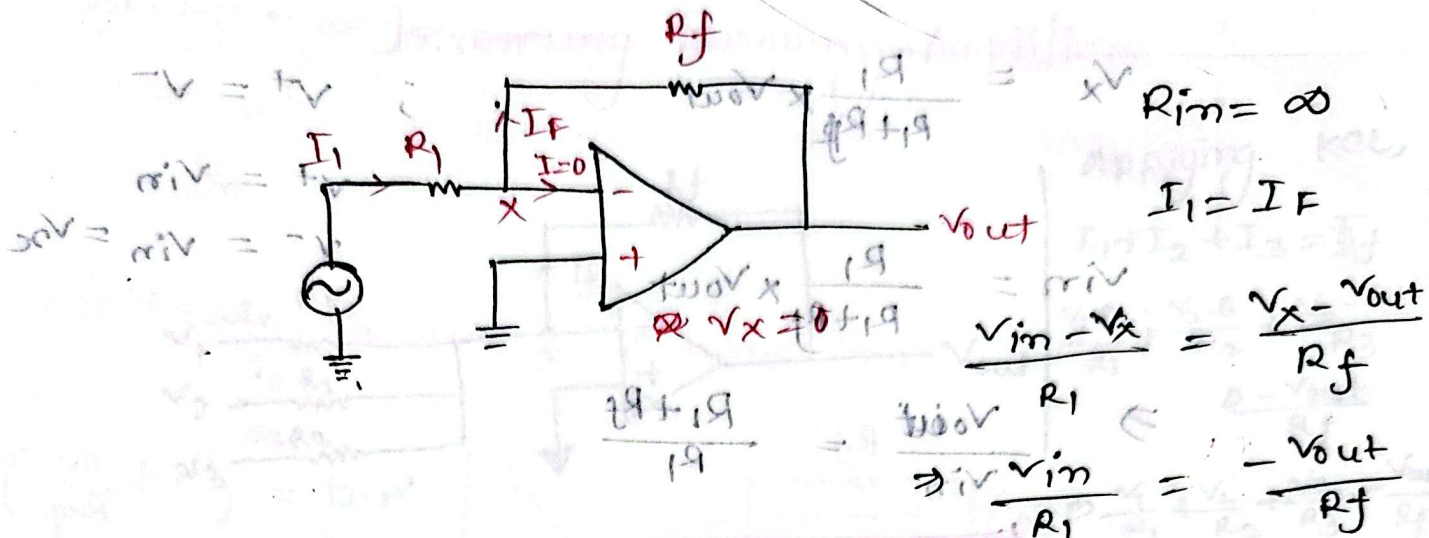
$$10 = 10^6 \times V_d$$

$$\Rightarrow V_d = 10 \mu V$$

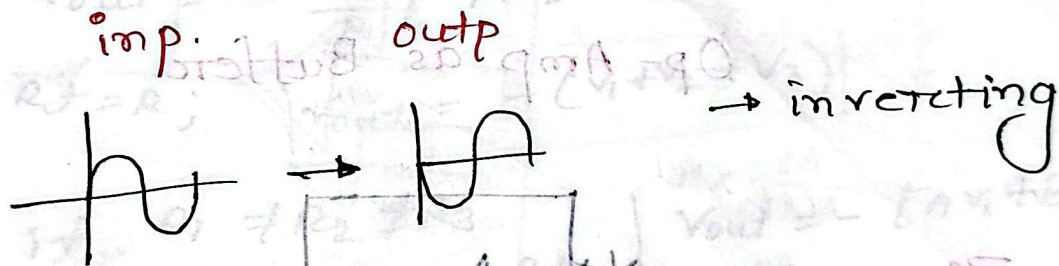
$$V^+ - V^- = 10 \mu V$$

$$V^+ - V^- \approx 0 V$$

$$V^+ = V^-$$

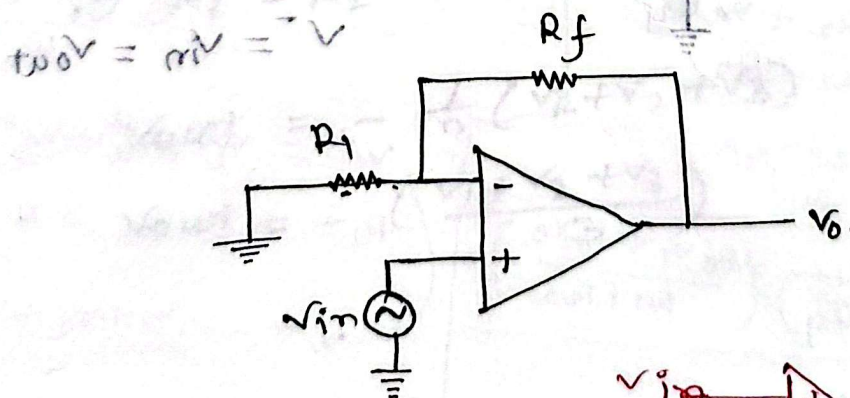


$$V_{out} = \left(\frac{V_{out}}{V_{in}} \right) = -\frac{R_f}{R_1} = A_{CL}$$

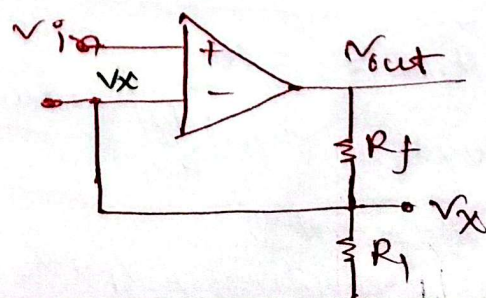


$\sigma = \frac{1}{2}$
 $\infty = 14$

Non-Inverting Amp.



redraw →



$\infty = \frac{V_x}{R_1 + R_f} \times V_{out}$; $V^+ = V^-$
 $V^+ = V_{in}$
 $V^- = V_{in} = V_{out}$

$V_{in} = \frac{R_1}{R_1 + R_f} \times V_{out}$

$\Rightarrow \frac{V_{out}}{V_{in}} = \frac{R_1 + R_f}{R_1}$

$A_v = \frac{V_{out}}{V_{in}} = \left(1 + \frac{R_f}{R_1}\right)$

Op-Amp as Buffer

$Z_{in} = \infty$

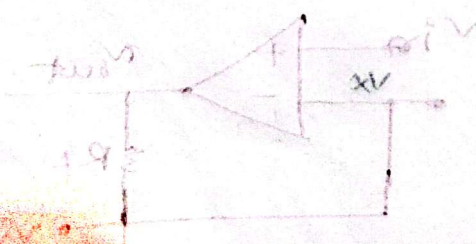
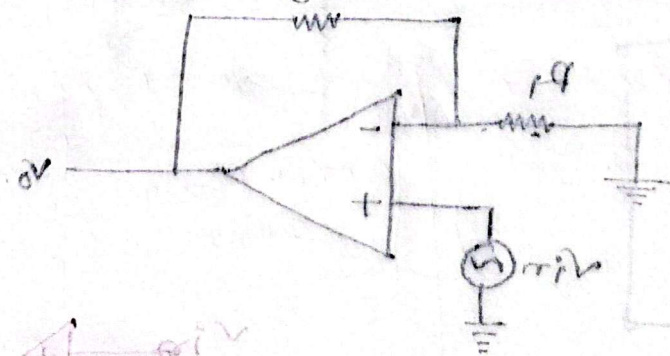
$V_{out} = V_{in}$

$R_f = 0$

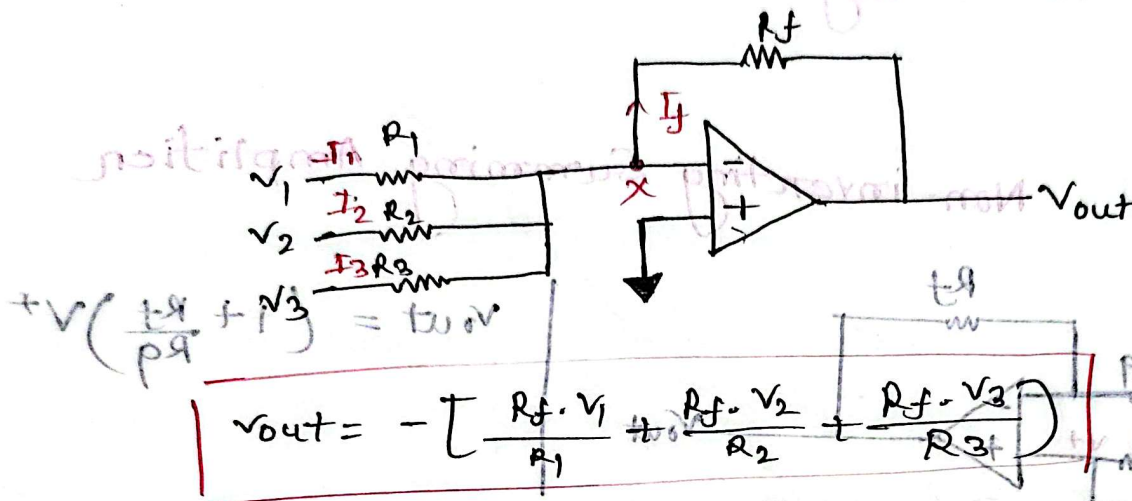
$R_1 = \infty$

$V^- = V^+$

$V^- = V_{in} = V_{out}$



Inverting Summing Amplifier



Applying KCL,

$$I_1 + I_2 + I_3 = I_f$$

$$\frac{V_1 - 0}{R_1} + \frac{V_2 - 0}{R_2} + \frac{V_3 - 0}{R_3} = \frac{0 - V_{out}}{R_f}$$

$$V_{out} = - \left[\frac{R_f \cdot V_1}{R_1} + \frac{R_f \cdot V_2}{R_2} + \frac{R_f \cdot V_3}{R_3} \right]$$

Let, $R_1 = R_2 = R_3 = R$

$$V_{out} = - \frac{R_f}{R} (V_1 + V_2 + V_3)$$

$R_f = R$;

$$V_{out} = - (V_1 + V_2 + V_3)$$

$$\frac{V_x \cdot \frac{1}{s} + \frac{1}{s}}{\frac{1}{s} + 1} = \frac{1}{s}$$

if $R_1 \neq R_2 \neq R_3$

$$\frac{R_f}{R_1} \neq \frac{R_f}{R_2} \neq \frac{R_f}{R_3}$$

$$V_{out} = - [A V_1 + B V_2 + C V_3]$$

$$\frac{1}{s} + V_{out} = \frac{1}{s} - \frac{1}{n} (V_1 + V_2 + V_3)$$

$$s = V_{out} \cdot s = - \frac{1}{n} (V_1 + V_2 + V_3)$$

$$\frac{s + s + 1}{s} = 1$$

$$1 + V \left(\frac{1}{s} + 1 \right) = 1$$

$$\frac{1 + s + 1}{s} \left(\frac{1}{s} + 1 \right) = 1$$

$$s = s = 1$$

$$\left(\frac{s + 1}{s} \right) \left(\frac{1}{s} + 1 \right) = 1$$

$$s = 1$$

Summing, Averaging, scaling, providing DC offset,

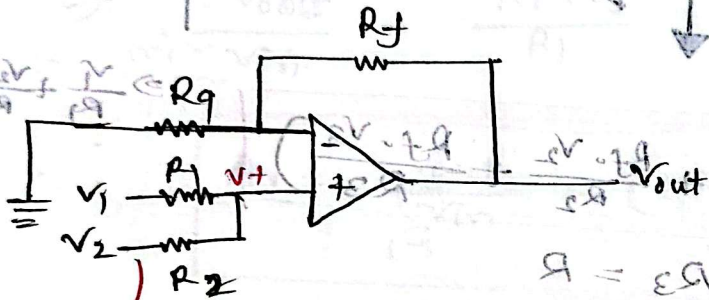
Digital to analog converter, Audio mixer.

$$I_{\Phi} = eI_1 + eI_2 + eI_3$$

$$\frac{0 - v_1}{R_1} + \frac{0 - v_2}{R_2} + \frac{0 - v_3}{R_3} = \frac{v_{out} - 0}{R_f}$$

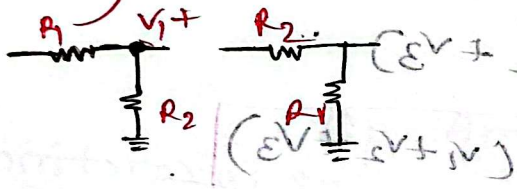
$$\frac{v_1}{R_1} + \frac{v_2}{R_2} + \frac{v_3}{R_3} = \frac{v_{out}}{R_f}$$

Non-inverting Summing Amplifier



$$v_{out} = \left(1 + \frac{R_f}{R_g}\right) V^+$$

$$R = R_1 = R_2 = R_3$$



$$V^+ = \frac{R_2}{R_1 + R_2} V_1 + \frac{R_1}{R_1 + R_2} V_2$$

$$V^+ = \frac{R_1}{R_1 + R_2} V_2$$

$$V^+ = \frac{V_1 R_2 + V_2 R_1}{R_1 + R_2}$$

$$V^+ = \frac{V_1 R_2 + V_2 R_1}{R_1 + R_2}$$

$$v_{out} = \left(1 + \frac{R_f}{R_g}\right) V^+$$

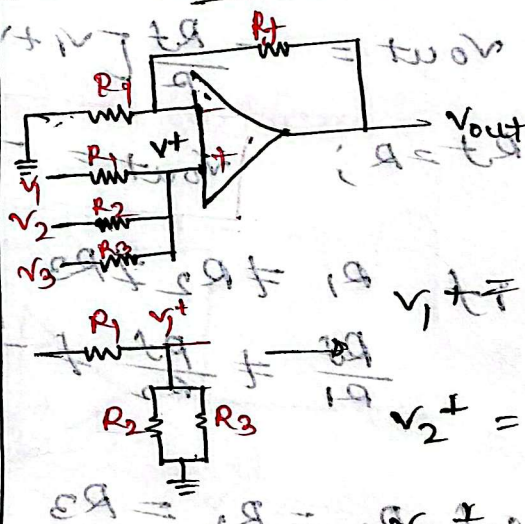
$$v_{out} = \left(1 + \frac{R_f}{R_g}\right) \left(\frac{R_2 V_1 + R_1 V_2}{R_1 + R_2}\right)$$

$$; R_1 = R_2 = R$$

$$v_{out} = \left(1 + \frac{R_f}{R_g}\right) \left(\frac{V_1 + V_2}{2}\right)$$

$$; R_f = R_g$$

$$v_{out} = V_1 + V_2$$



$$R = R_1 = R_2 = R_3$$

$$V^+ = \frac{V_1 + V_2 + V_3}{3}$$

$$R_1 = R_2 = R_3 = R$$

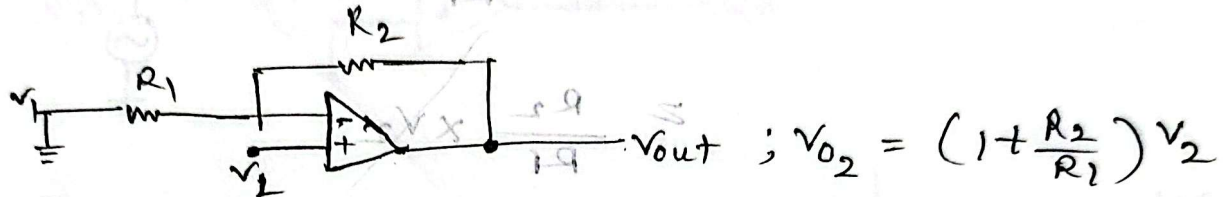
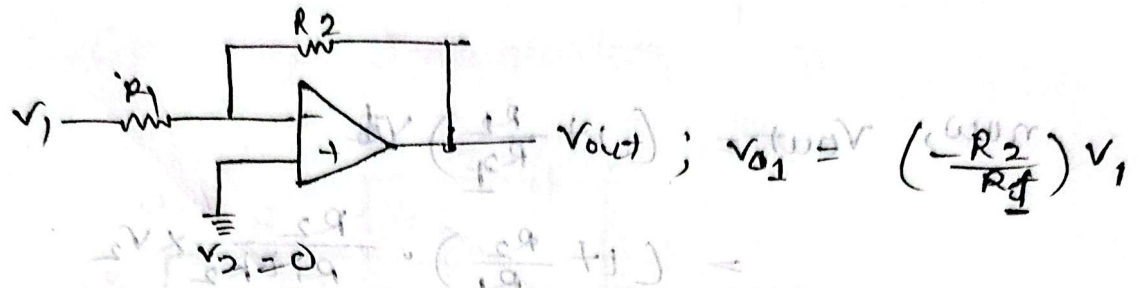
$$V^+ = \frac{V_1 + V_2 + V_3}{3}$$

$$v_{out} = \left(1 + \frac{R_f}{R_g}\right) V^+$$

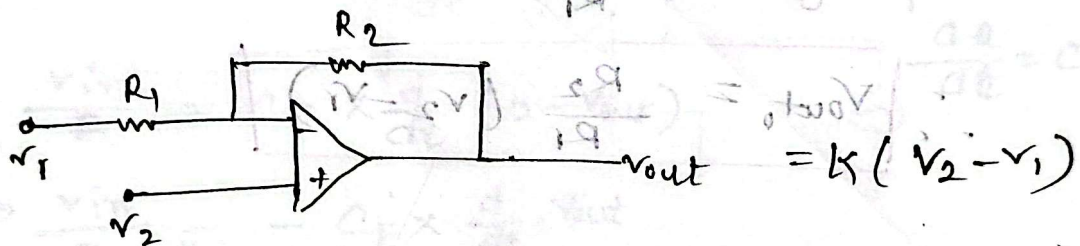
$$v_{out} = \left(1 + \frac{R_f}{R_g}\right) \left(\frac{V_1 + V_2 + V_3}{3}\right)$$

$$R_f = R_g = R$$

Op-Amps as Differential Amplifiers



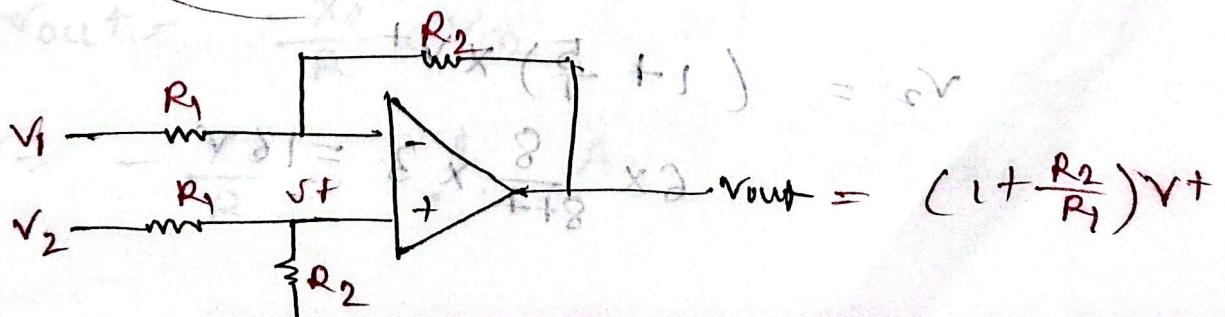
$$V_{out} = V_{01} + V_{02} = \left(-\frac{R_2}{R_1}\right) V_1 + \left(1 + \frac{R_2}{R_1}\right) V_2$$



and V_{out}

$$\Rightarrow \frac{1 + \frac{R_2}{R_1}}{\frac{R_1 + R_2}{R_1}} = \frac{\frac{R_2}{R_1}}{\frac{R_2}{R_1}} = 1$$

$$\Rightarrow \left(\frac{R_2}{R_1 + R_2} \right) \times \frac{R_1 + R_2}{R_1} = \frac{R_2}{R_1} = 1$$



$$v_1 = \frac{R_2}{R_1 + R_2} \times V_2$$

$$v_1(\text{now}), V_{out} = \left(1 + \frac{R_2}{R_1}\right) v_1$$

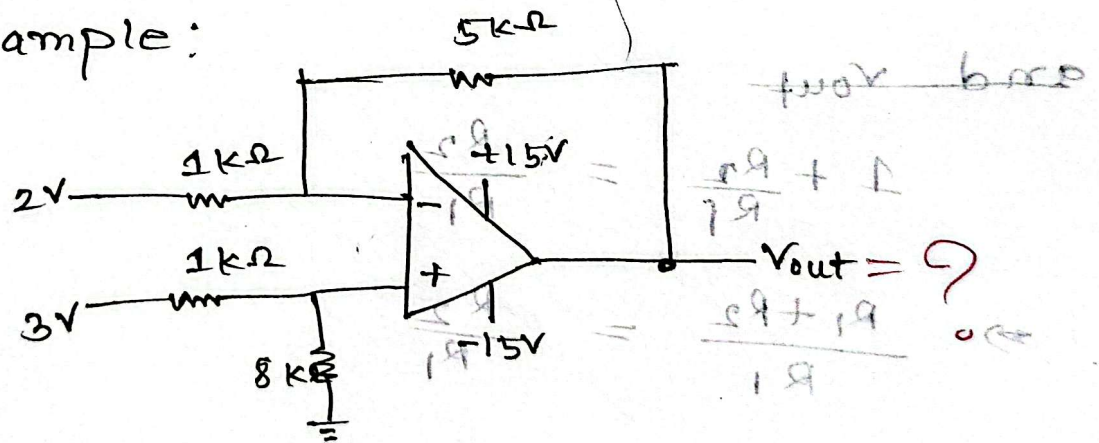
$$= \left(1 + \frac{R_2}{R_1}\right) \cdot \frac{R_2}{R_1 + R_2} \times V_2$$

$$v_1 \left(\frac{R_2}{R_1} + 1 \right) = v_{out} \Rightarrow \frac{R_2}{R_1} \times V_2$$

$$v_1 \left(\frac{R_2}{R_1} + 1 \right) + v_1 \left(-\frac{R_2}{R_1} \right) = v_{out} \Rightarrow \frac{R_2}{R_1} V_2$$

$$V_{out} = \frac{R_2}{R_1} (V_2 - V_1)$$

Example:



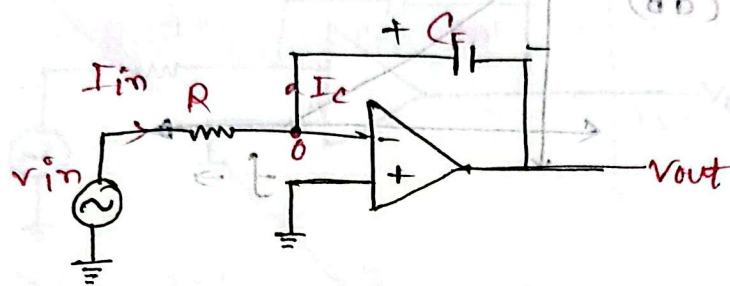
$$V_{01} = \frac{R_2}{R_1} \left(-\frac{5}{1} \right) \times 2 = -10V$$

$$V_2 = \left(1 + \frac{5}{1}\right) \times V_1$$

$$v_1 \left(\frac{R_2}{R_1} + 1 \right) = v_{out} \Rightarrow 6 \times \frac{8}{8+1} \times 3 = 16V$$

$$V_{out} = V_{02} + V_{01} = 16 - 10 = 6V$$

Op-Amp Integrator



$$I_n = I_c$$

$$\Rightarrow \frac{v_{in}}{R} = C_F \times \frac{dV_c}{dt}$$

$$I_c = C \cdot \frac{dV_c}{dt}$$

$$Q = C V_c$$

$$\frac{dQ}{dt} = C \cdot \frac{dV_c}{dt} = I_c$$

$$\Rightarrow \frac{v_{in}}{R} = C_F \times \frac{d}{dt} (0 - V_{out})$$

$$\Rightarrow \frac{v_{in}}{R} = -C_F \times \frac{d}{dt} V_{out}$$

$$\Rightarrow \frac{d}{dt} (V_{out}) = -\frac{1}{R \times C_F} \times V_{in}(t)$$

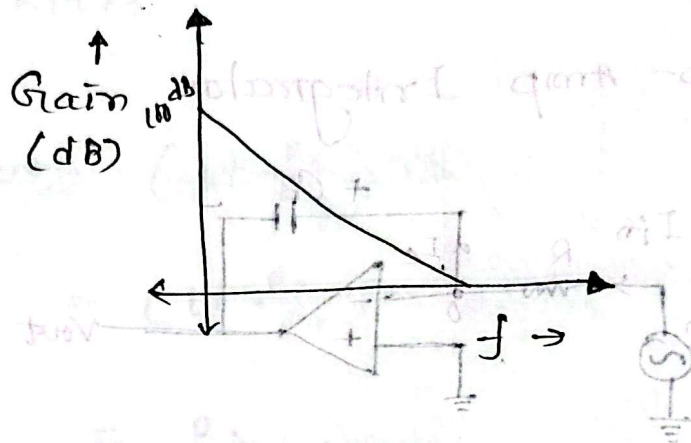
$$V_{out} = -\frac{1}{R \cdot C_F} \int V_{in}(t) \cdot dt$$

$$X_c = \frac{1}{\omega C} = \frac{1}{2\pi f C}$$

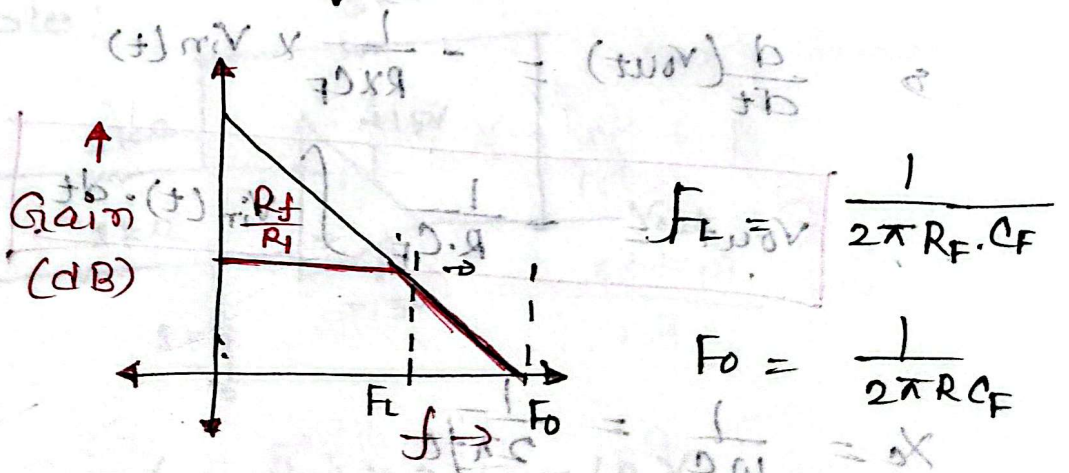
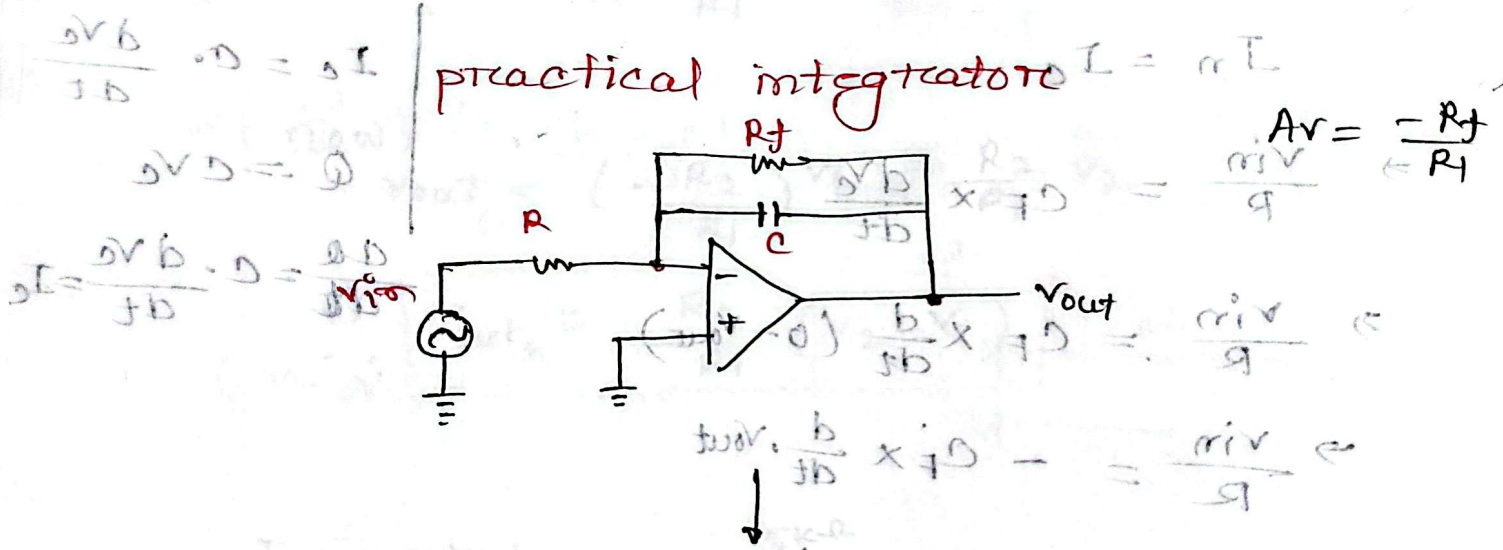
$$V_{out} = -\frac{X_c}{R} \times V_{in}$$

$$A_v = -\frac{X_c}{R} = -\frac{1}{2\pi f C R}$$

Frequency response of ideal Integrator



practical integrator



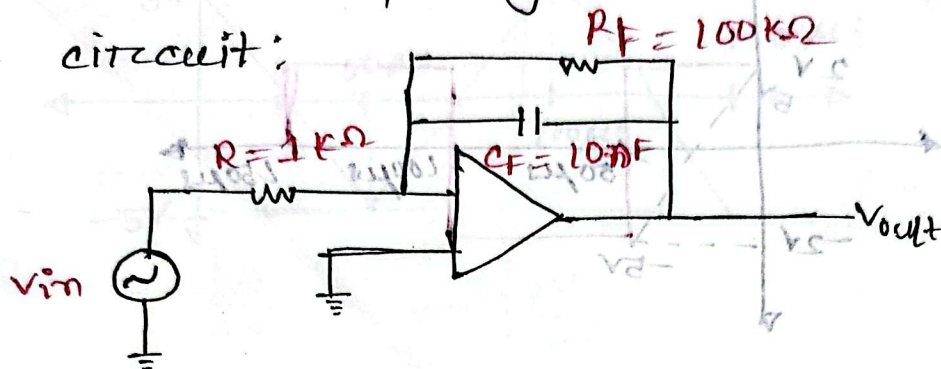
$$F_s > 10 F_L$$

Signal frequency

Cutoff frequency

Example 2) integrator

Find lower frequency limit of integration for the circuit:



$$f_L = \frac{1}{2\pi R_F C_F} = 159 \text{ Hz} \quad ; \quad \boxed{f_L = 159 \text{ Hz}}$$

$$\text{So, } f_s > 10 \times f_L = 1590 \text{ Hz} = 1.59 \text{ kHz}$$

$$v_{in} = \sin(2\pi \times 5000t)$$

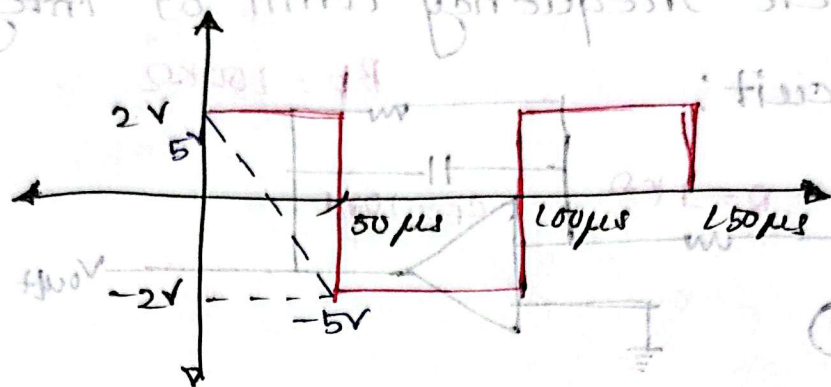
$$v_{out} = -\frac{1}{R C_F} \times \int v_{in}(t) \cdot dt$$

$$= -\frac{1}{1 \times 10^3 \times 10 \times 10^{-9}} \int \sin(2\pi \times 5000t) \cdot dt$$

$$= \frac{10^5}{2\pi \times 5000} \times \cos(2\pi \times 5000t)$$

$$\therefore v_{out} = 3.18 \cos(2\pi \times 5000t)$$

Example (same circuit):



$$C = 100 \mu F, f = 10 \text{ kHz}$$

$$f_L = 159 \text{ Hz}$$

$$f_s = 10 \text{ kHz}$$

now; $v_1(t) = 2V; 0 < t < 50$

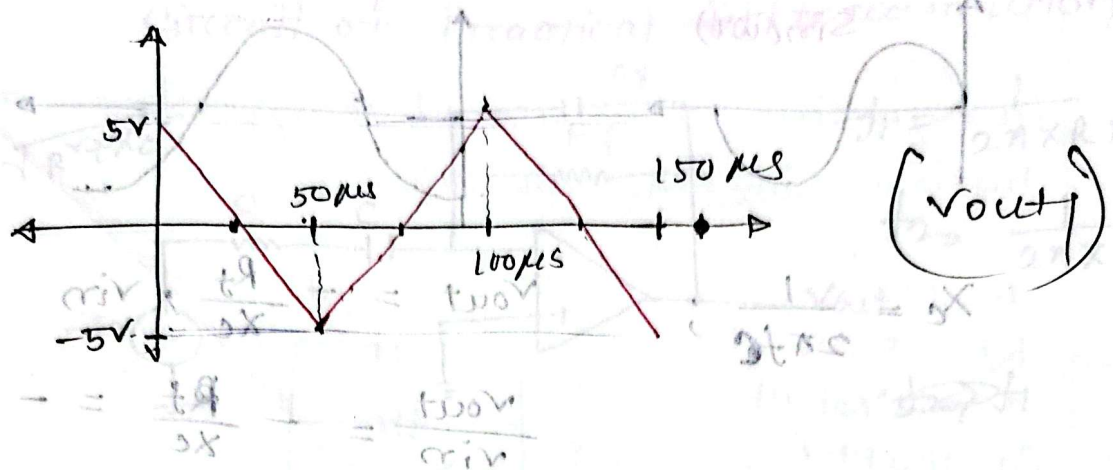
$$v_2(t) = -2V; 50 < t < 100$$

$$V_{01} = -\frac{1}{RC} \int_0^{50 \mu s} v_1(t) \cdot dt$$

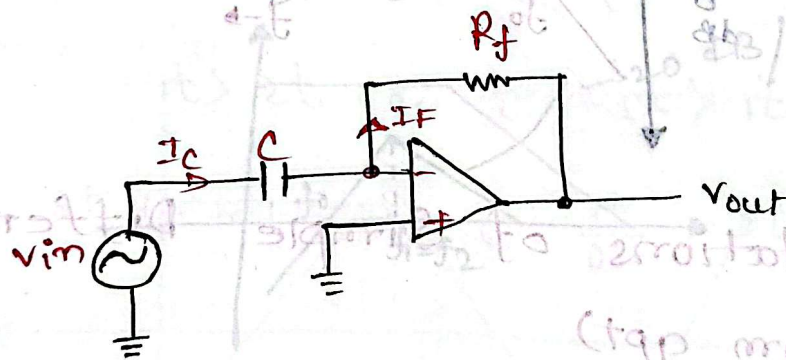
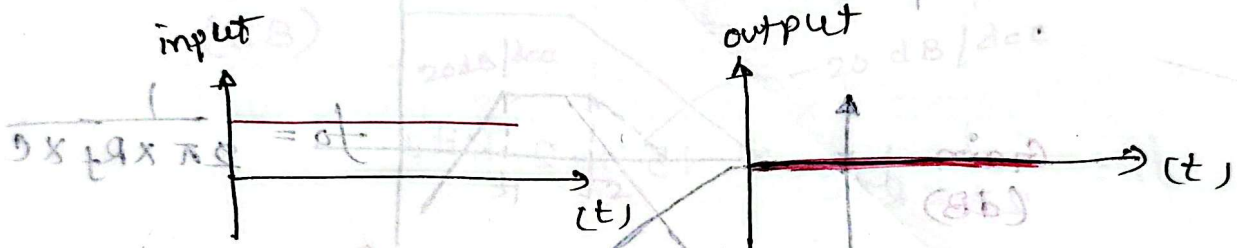
$$= -10^5 \times 50 \mu s \times 2$$

$$V_{01} = -10V$$

$$V_{02} = -10V$$



Op-Amp as Differentiator

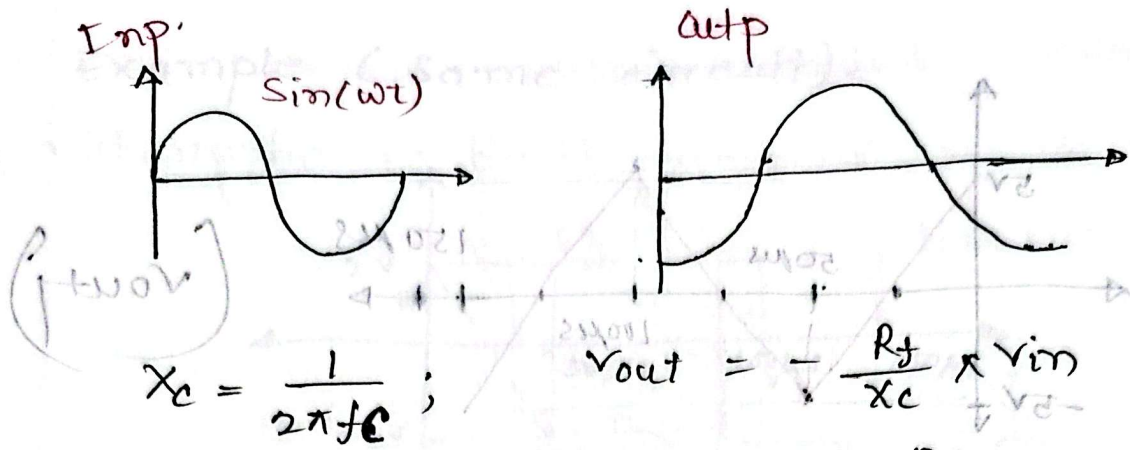


$$I_C = I_F$$

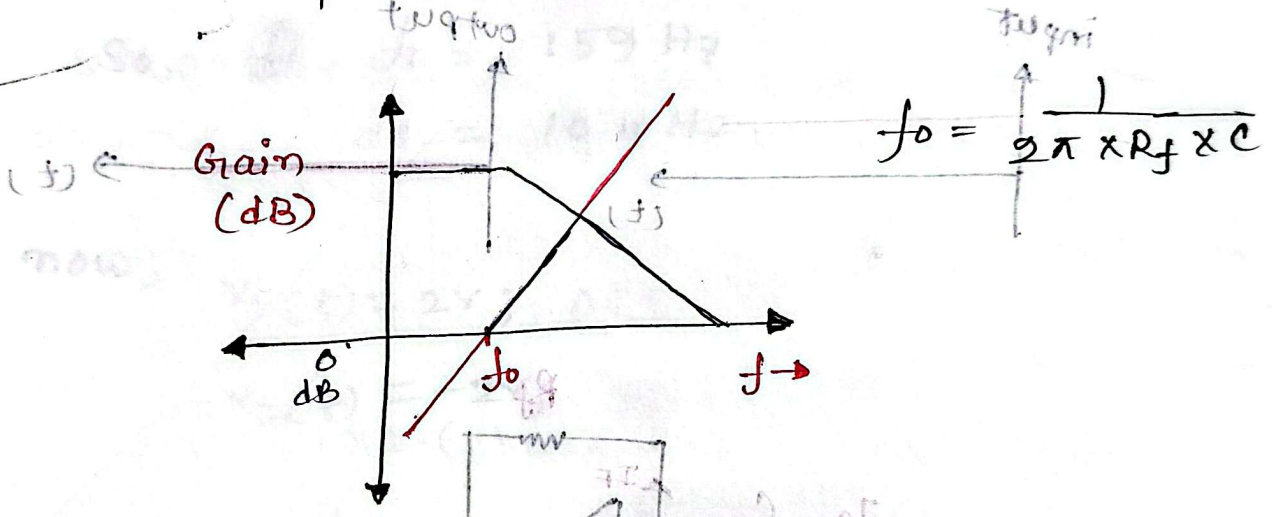
$$\Rightarrow C \cdot \frac{dv_c}{dt} = \frac{0 - v_{out}}{R_f}$$

$$\Rightarrow C \cdot \frac{dv_{in}}{dt} = -\frac{v_{out}}{R_f}$$

$$\Rightarrow \boxed{v_{out} = -C \cdot R_f \cdot \frac{dv_{in}}{dt}}$$



Gain = $\left| \frac{V_{out}}{V_{in}} \right| = 2\pi \times R_f \times f \times C$



* Limitations of Simple Differentiator (from ppt)

$I = \frac{V}{R}$

$\frac{V_{out} - 0}{R_f} = \frac{V_{in}}{R_f} \cdot \frac{1}{f} \cdot \frac{1}{C}$

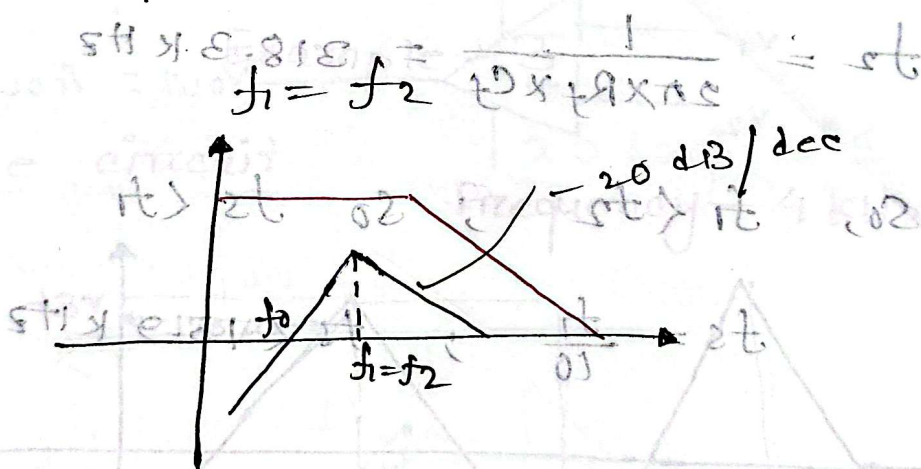
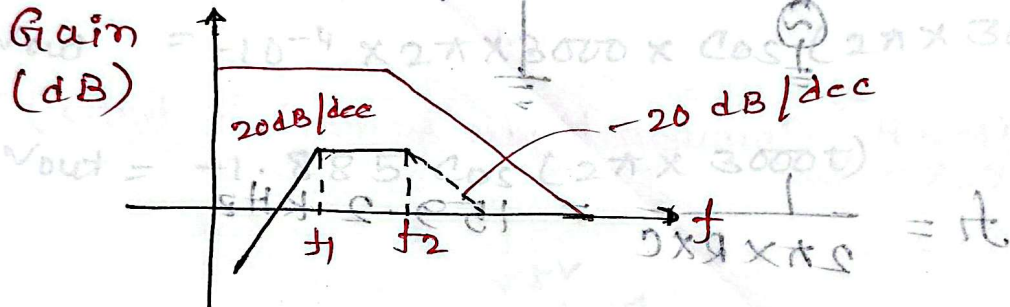
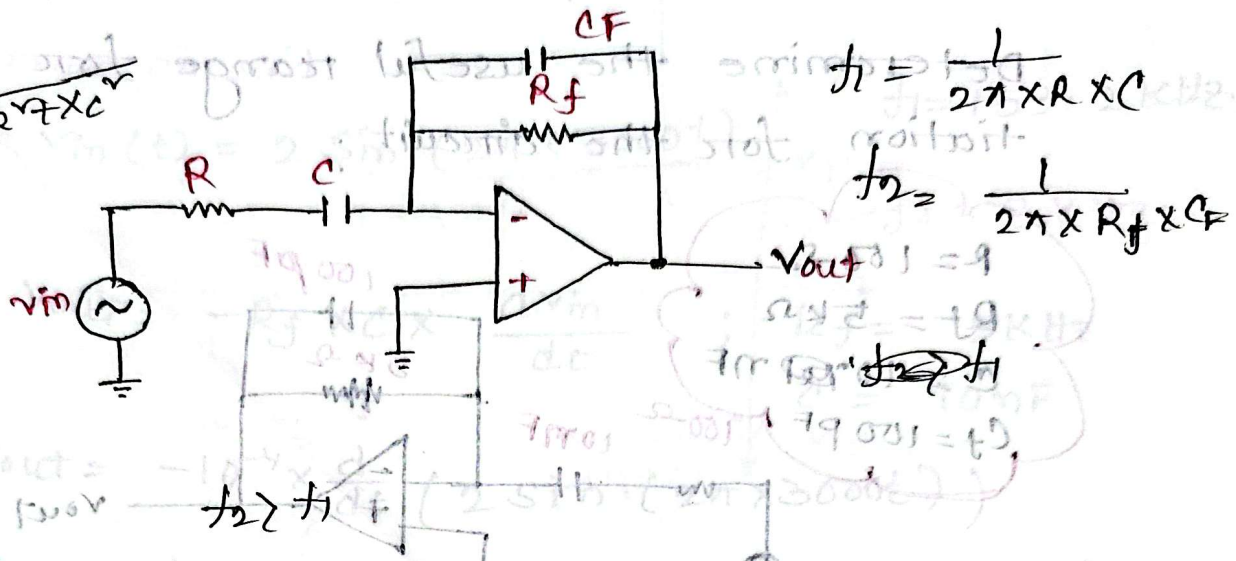
$\frac{V_{out}}{R_f} = \frac{V_{in}}{R_f} \cdot \frac{1}{f} \cdot \frac{1}{C}$

$\frac{V_{out}}{f} = \frac{V_{in}}{f} \cdot \frac{1}{C}$

$V_{out} = V_{in} \cdot \frac{1}{C} \cdot \frac{1}{f}$

Circuit of Practical Differentiator

$$Z_{in} = \sqrt{R^2 + X_C^2}$$



$$\text{slope} = \frac{6V}{125 \mu s} = 48000 \text{ V/s}$$

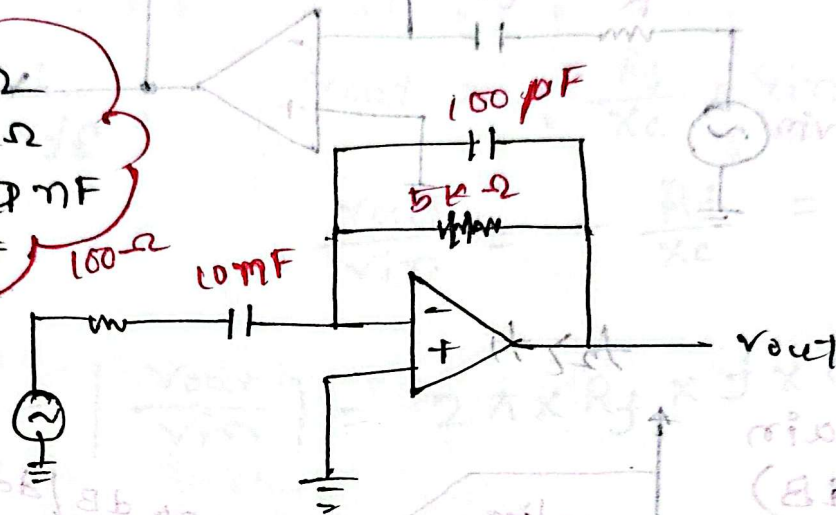
$$V_{out} = -R_f \times C \times \frac{dV_{in}}{dt} = 5k\Omega \times 10 \text{ nF} \times 48000$$

$$= -2.4V$$

Example

Determine the useful range for differentiation for the circuit:

$$\begin{aligned} R &= 100 \Omega \\ R_f &= 5 k\Omega \\ C &= 10 \mu F \\ C_f &= 100 pF \end{aligned}$$



$$f_1 = \frac{1}{2\pi \times R \times C} = 159.2 \text{ kHz}$$

$$f_2 = \frac{1}{2\pi \times R_f \times C_f} = 318.3 \text{ kHz}$$

$$\text{So, } f_1 < f_2 ; \text{ So } f_s < f_1$$

$$f_s = \frac{f_1}{10} ; f_s < 15.9 \text{ kHz}$$

Example - 2

→ Same circuit

$$v_{in}(t) = 2 \sin(2\pi \times 3000t)$$

$$f_1 = 159.2 \text{ kHz}$$

$$f_s = 3 \text{ kHz}$$

$$v_{out} = -R_f \times C \times \frac{dv_{in}}{dt} ; R_f = 5 \text{ kHz}$$

$$C = 10 \text{ nF}$$

$$\therefore v_{out} = -10^{-4} \times \frac{d}{dt} (2 \sin(2\pi \times 3000t))$$

$$v_{out} = -10^{-4} \times 2\pi \times 3000 \times \cos(2\pi \times 3000t)$$

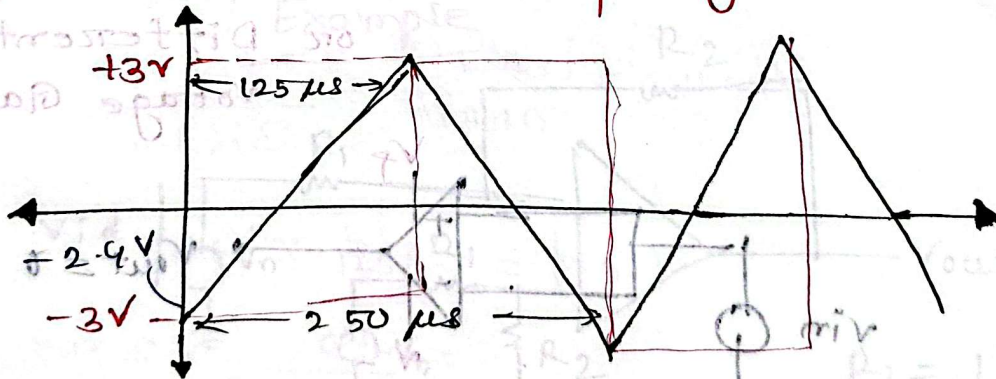
$$v_{out} = -1.885 \cos(2\pi \times 3000t)$$

$$(v - V) \times 50 \text{ A} = 10 \text{ V}$$

Example - 3

→ Same circuit

$$\text{Frequency} = 4 \text{ kHz} \quad T = 250 \mu\text{s}$$

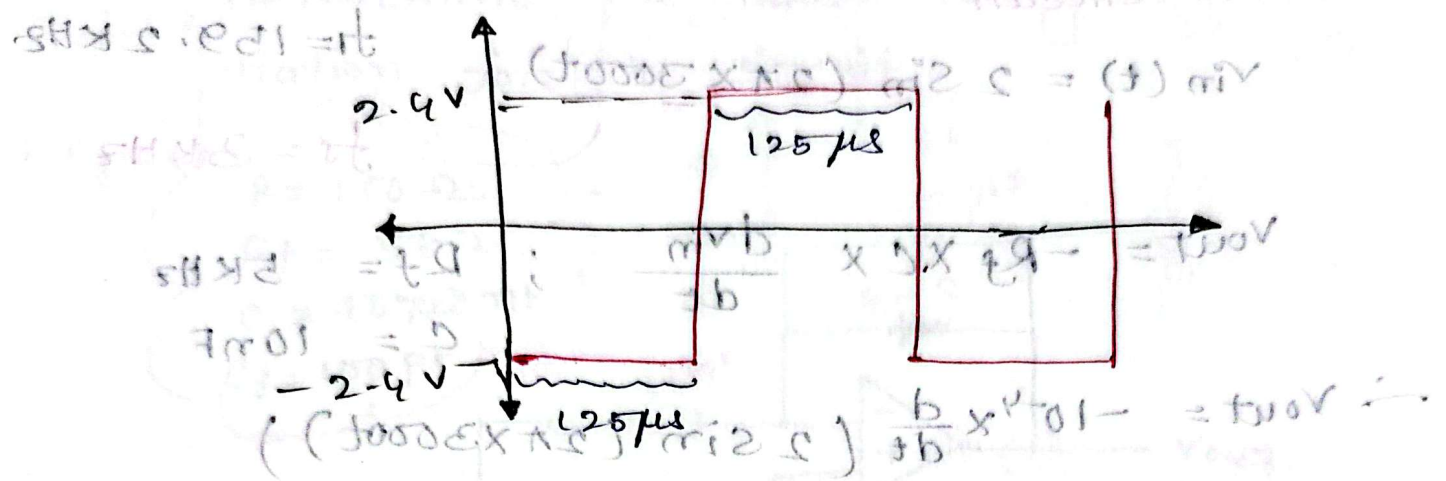


$$\text{slope} = \frac{6 \text{ V}}{125 \mu\text{s}} = 48000 \text{ V/s}$$

$$v_{out} = -R_f \times C \times \frac{dv_{in}}{dt} = 5 \text{ k}\Omega \times 10 \text{ nF} \times 48000$$

$$= -2.4 \text{ V}$$

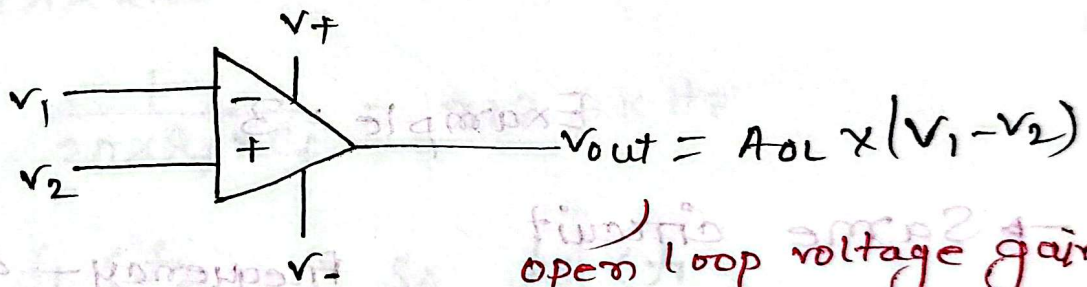
output :



$$(10000 \times \text{K}\Omega) \times 200 \times 10^{-6} \times 10^{-1} = 200V$$

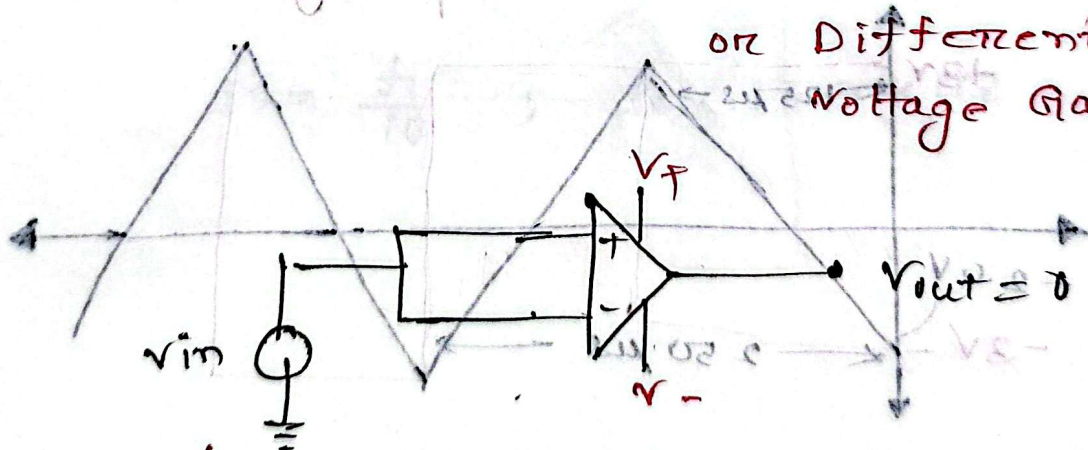
CMRR (Common Mode Rejection Ratio):

$$(10000 \times \text{K}\Omega) \times 200 \times 10^{-6} \times 10^{-1} = 200V$$



open loop voltage gain

or Differential Voltage Gain

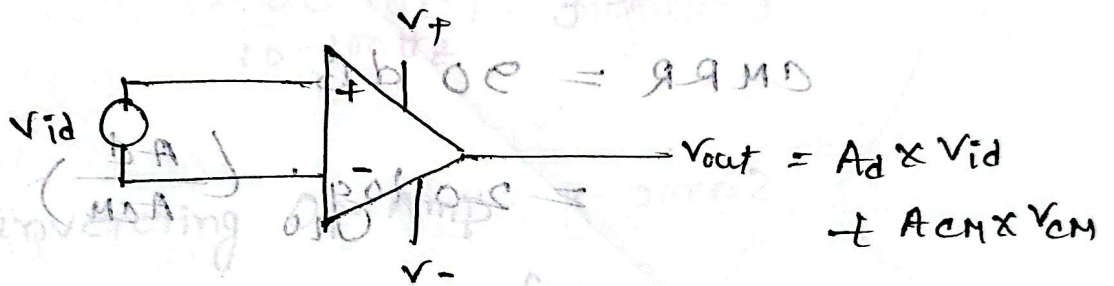
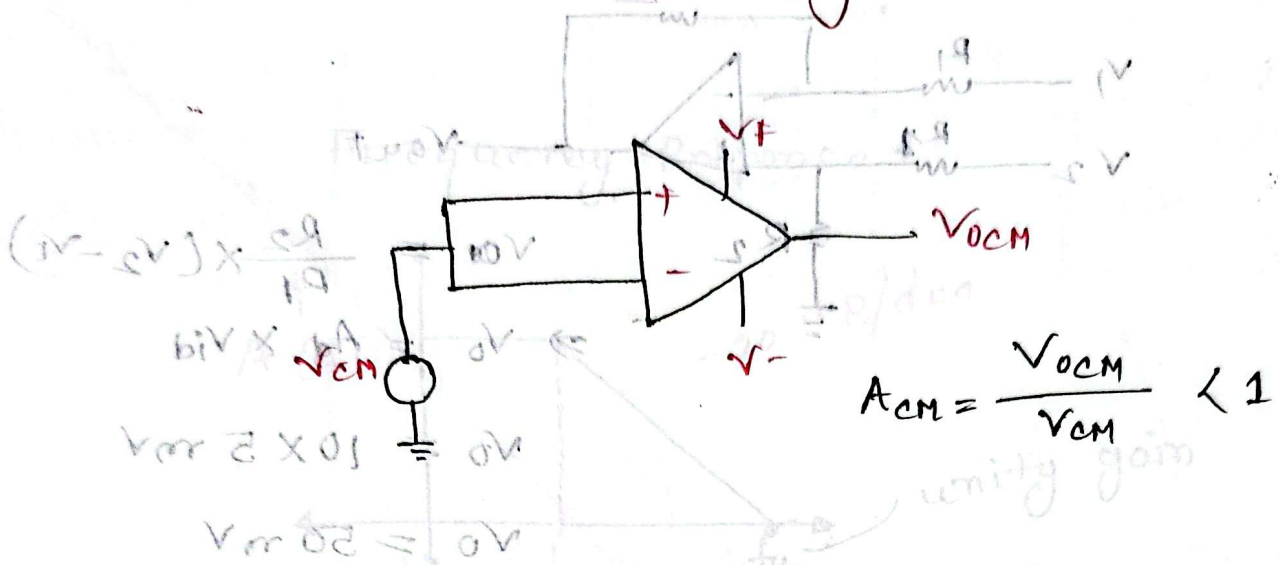


Common mode configuration

$$10000 \mu \times 70 \mu \times 10^{-6} = 2.5 \times 10^{-6} \times 10^{-6} = 2.5 \times 10^{-12}$$

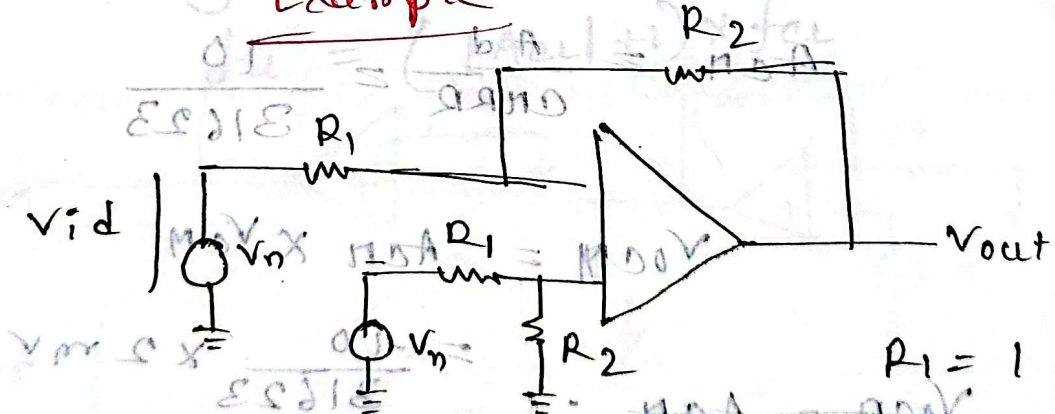
$$V_{in} = 0.1V$$

Common Mode Configuration



CMRR = $\frac{A_d}{A_{CM}} = 20 \log \left(\frac{A_d}{A_{CM}} \right)$

Example



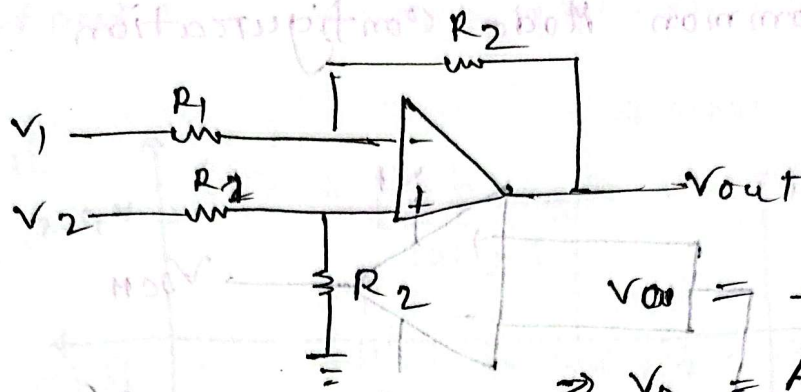
$R_1 = 1 \text{ k}\Omega$

$R_2 = 10 \text{ k}\Omega$

CMRR = 90 dB

$V_{id} = 5 \text{ mV}$ at 1 kHz

$V_{CM} = 2 \text{ mV}$ at 50 Hz



$$V_{out} = \frac{R_2}{R_1} \times (V_2 - V_1)$$

$$\Rightarrow V_o = A_d \times V_{id}$$

$$V_o = 10 \times 5 \text{ mV}$$

$$V_o = 50 \text{ mV}$$

$$\text{CMRR} = 90 \text{ dB}$$

$$= 20 \log_{10} \left(\frac{A_d}{A_{CM}} \right)$$

$$\Rightarrow \frac{A_d}{A_{CM}} = 10^{4.5} \approx 31623$$

$$\text{CMRR} = 31623 = \frac{A_d}{A_{CM}}$$

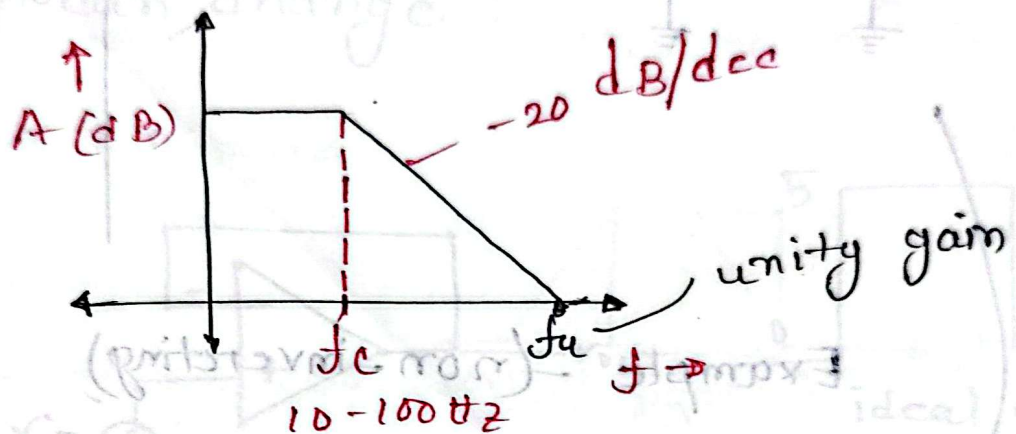
$$A_{CM} = \frac{A_d}{\text{CMRR}} = \frac{10}{31623}$$

$$V_{OCM} = A_{CM} \times V_{CM}$$

$$V_{OCM} = 0.63 \mu\text{V}$$

$E = (\text{Gain} \times \text{Bandwidth}) \text{ Product}$

Frequency Response



Non-inverting Op-Amp

$$f_u = A_{cl} \times f_{cl}$$

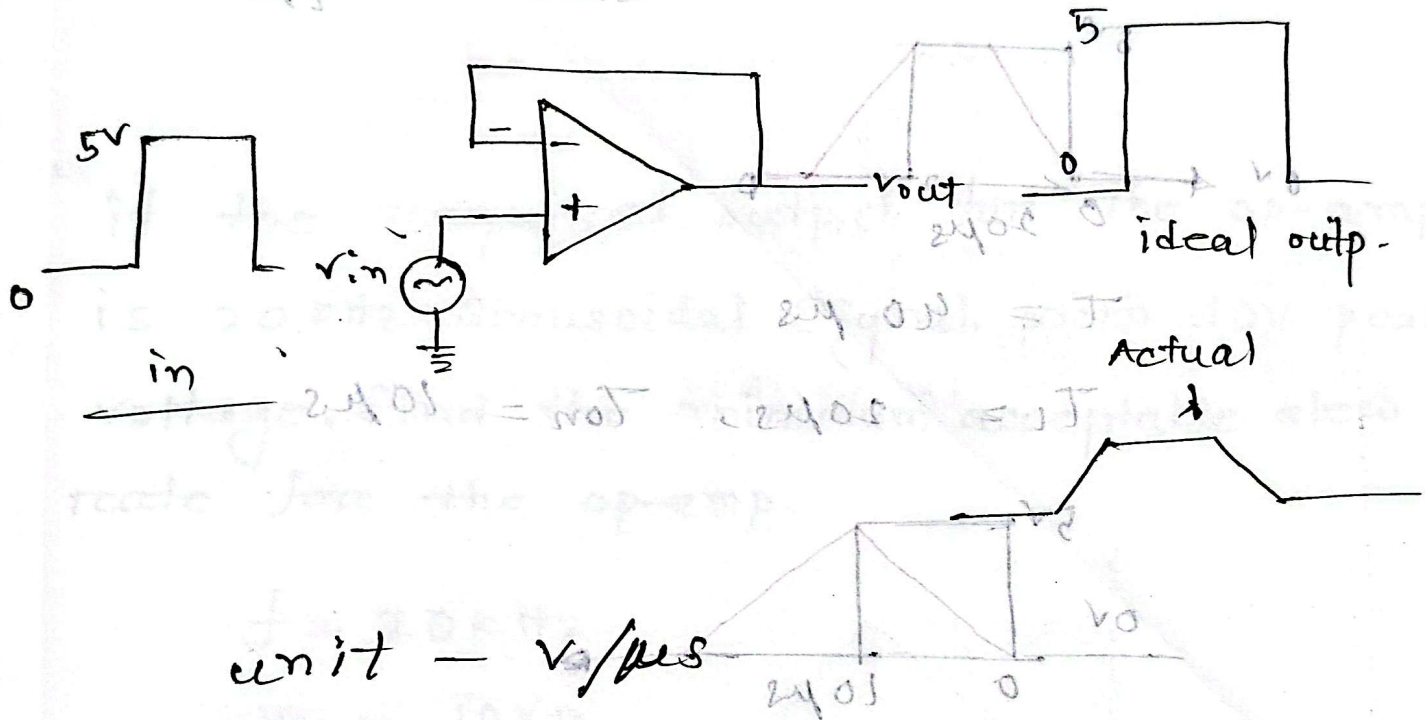
Inverting

$$f_u = (|A_{cl}| + 1) \times f_{cl}$$

Op-amp Slew Rate

The maximum rate at which the output of op-amp can change.

$$2.5 \text{ V} / 3.0 = 0.83$$



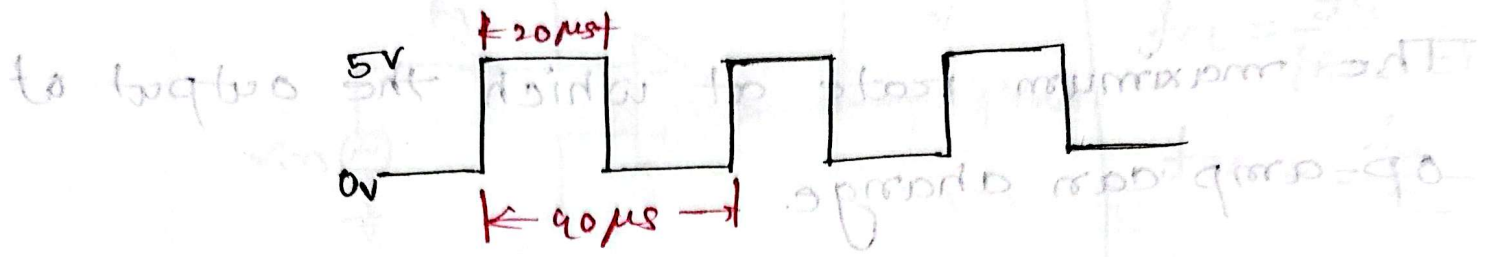
Internal Compensation Capacitor; C_c

$$i_c = C_c \frac{dv_c}{dt}$$

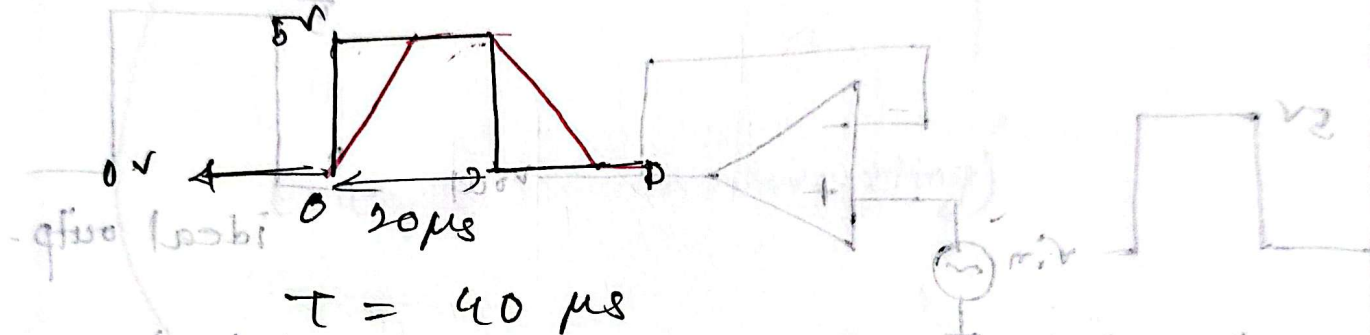
$$\frac{dv_c}{dt} = \frac{i_c}{C_c} = \frac{200 \mu\text{A}}{300 \text{ pF}} = 6.66 \text{ V}/\mu\text{s}$$

$$\left[\frac{v_b}{f \times R_c \times mV} = \left| \frac{v_b}{f \times R_c} \right| \right] = 6.66 \text{ V}/\mu\text{s}$$

Effect of slew rate on Pulse Signal

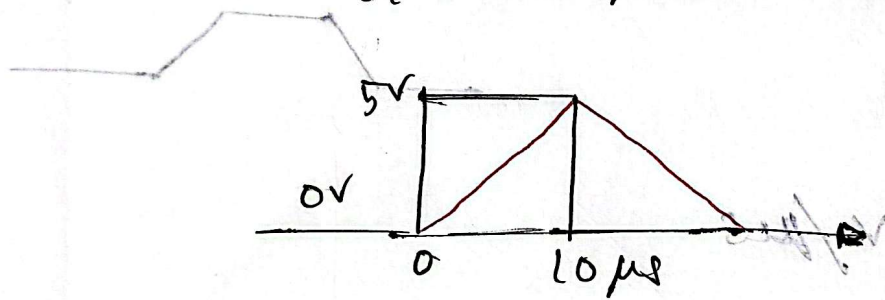


$$SR = 0.5 \text{ V}/\mu s$$

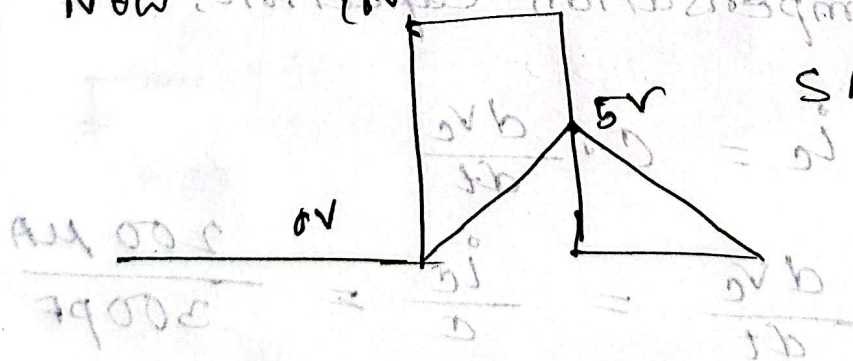


$$\tau = 40 \mu s$$

$$T_i = 20 \mu s, T_{on} = 10 \mu s$$



Now:



$$SR = 0.5 \text{ V}/\mu s$$

$$\left| \frac{dv}{dt} \right|_{\max} = V_m \times 2\pi \times f$$

$$S.R = V_m \times 2\pi \times f$$

$$f_{max} = \frac{SR}{2\pi \times V_m}$$

Ex - 1

if the required output for the op-amp is 20 kHz sinusoidal signal, with 10V peak voltage. Find the minimum acceptable slew rate for the op-amp.

$$f = 20 \text{ kHz}$$

$$V_m = 10 \text{ V}$$

$$SR = V_m \times 2\pi \times f$$

$$= 10 \times 2\pi \times 20000 = 1.257 \text{ MV/s}$$

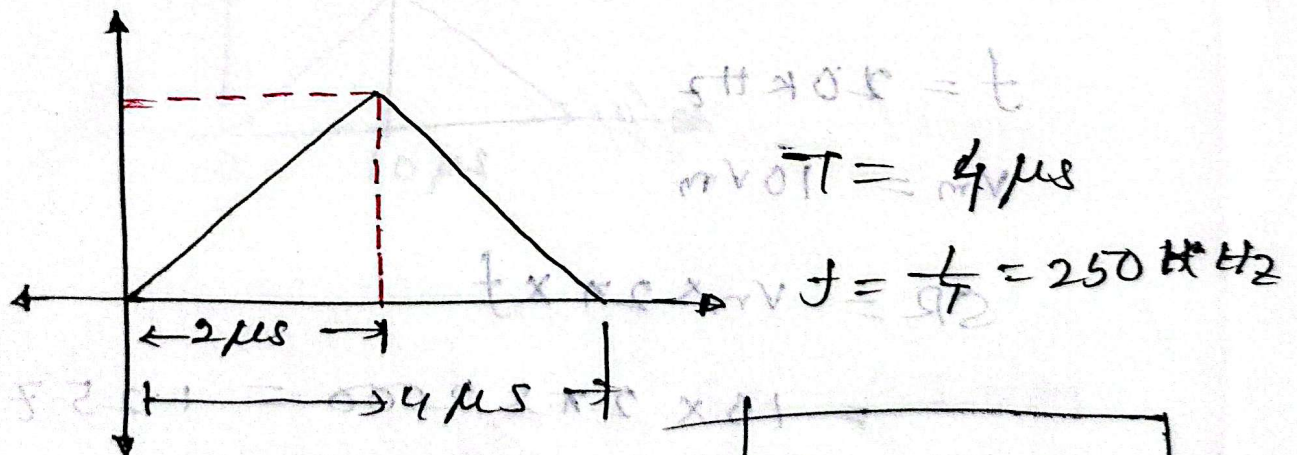
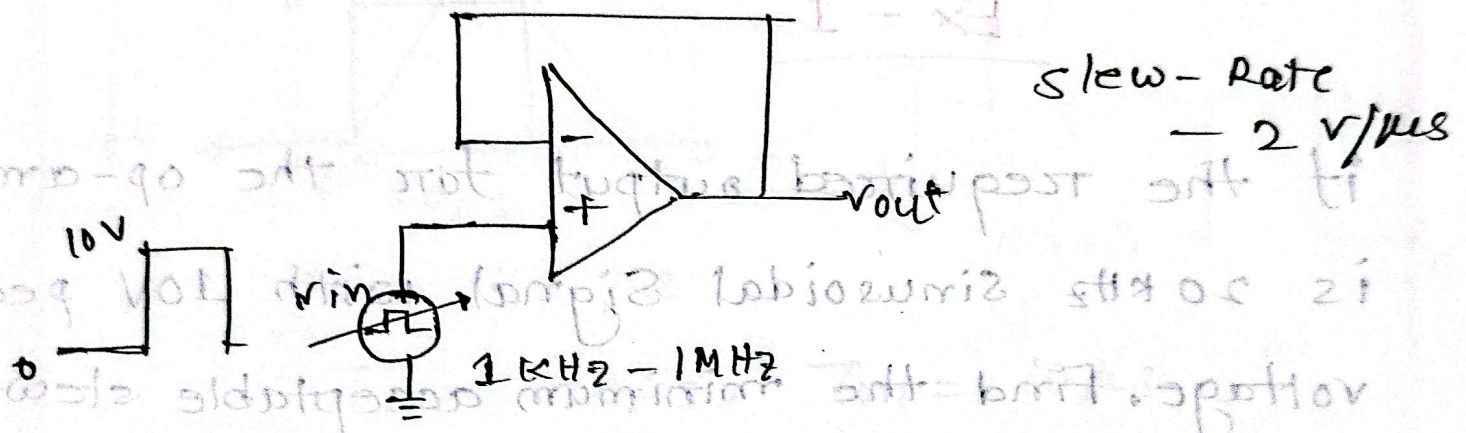
$$= 1.257 \text{ V/}\mu\text{s}$$

$$20 \text{ kHz} = 20000 \text{ Hz}$$

$$10 \text{ V} = 10 \text{ V}$$

Ex-2

Find the frequency of the square wave pulse, where the output will be triangular wave with peak voltage of 4V



$$T_1 = 10 \mu s$$

$$f_1 = 100 \text{ kHz}$$